Lecture Notes 5b
Consumption and Investment

Intermediate Macroeconomics, EC2211

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Contents and literature

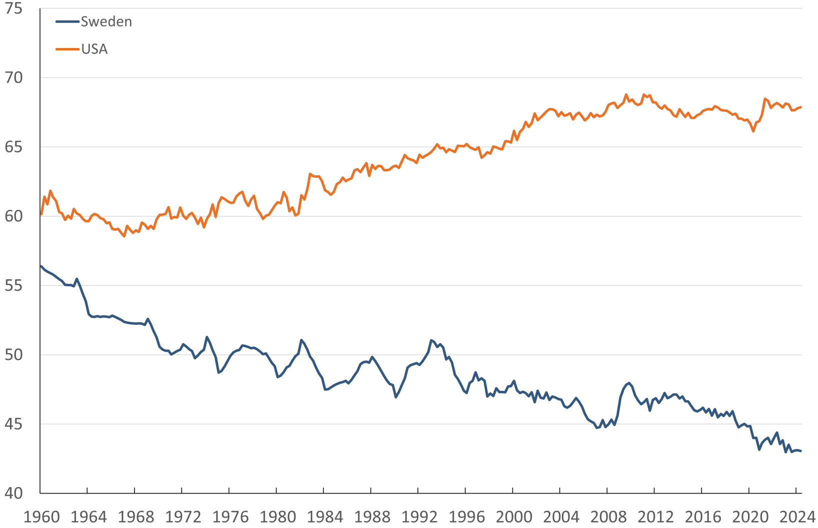
- Consumption
- Investment

Literature:

- Jones (2024), ch. 16
- Jones (2024), ch. 17.1 - 17.2 and box on Tobin's q in ch. 17.3

Consumption

The consumption share in output

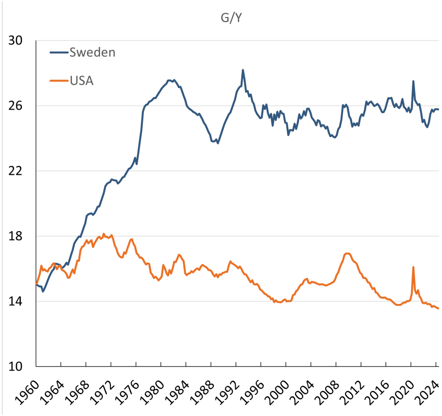


Private final consumption expenditure, percent of GDP, current prices. Source: OECD EO.

Consumption

Consumption accounts for a large fraction of GDP

Why has C/Y trended down in Sweden and up in the United States?



Government expenditure (% of GDP). Source: OECD EO.

- Sweden: more government consumption
- U.S.: less investment and net exports

Consumption theories

Neoclassical theory

- Rational households
- Forward-looking expectations
- Well-functioning markets
- Consumption and savings chosen to maximize life-time utility

Alternatives

- Keynesian theory: ad-hoc assumptions
- Behavioral models: based on evidence from psychology etc
- Models with various frictions and imperfections, e.g. borrowing constraints

A two-period neoclassical model of consumption

Notation:

- y_t: disposable income in period t, $t = 1, 2$
- c_t: consumption in period t
- s: saving in period 1

The consumer faces the following two budget constraints:

$$c_1 = y_1 - s \tag{1}$$

$$c_2 = y_2 + (1 + r)s \tag{2}$$

The intertemporal budget constraint

Combining (1) and (2) yields the consumer's intertemporal budget constraint (IBC):

$$c_1 + \frac{c_2}{1+r} = y_1 + \frac{y_2}{1+r} \tag{3}$$

Left side of (3): present value of lifetime expenditure (consumption)

Right side of (3): present value of lifetime income

The utility function

The consumer derives utility, U, from consumption according to:

$$U = u(c_1) + \beta u(c_2) \tag{4}$$

where $\beta \in [0, 1]$ is the discount factor, measuring how the consumer values the future relative to today (patience).

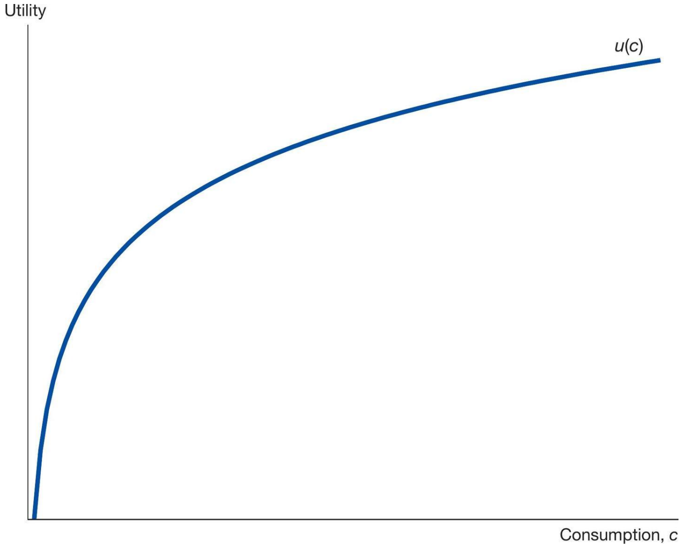
The flow utility function $u(c_t)$ has the following properties:

$$u'(c_t) > 0 \tag{5}$$

$$u''(c_t) < 0 \tag{6}$$

Assumptions (5) and (6) imply that utility is increasing in consumption, but that there is *diminishing marginal utility*

Figure 16.1



From Jones (2024)

The consumer's maximization problem

The consumer faces the following problem:

$$\max_{c_1, c_2} U = u(c_1) + \beta u(c_2) \tag{7}$$

subject to

$$c_1 + \frac{c_2}{1+r} = y_1 + \frac{y_2}{1+r}$$

Two ways of solving the consumer's problem:

1. Substitution
2. Setting up the Lagrangian

Solving by substituting for c_2

Substitute for c_2 from the IBC into the utility function

Then a maximization problem in one variable:

$$\max_{c_1} u(c_1) + \beta u \left[\underbrace{(1+r)(y_1 - c_1) + y_2}_{c_2} \right]$$

The FOC is:

$$u'(c_1) + \beta u'(c_2) \left[-(1+r) \right] = 0 \tag{8}$$

The Euler equation

Simplifying (8) gives the **Euler equation**:

$$u'(c_1) = \beta(1 + r)u'(c_2) \tag{9}$$

When consumption is chosen optimally, the consumer must be indifferent between:

- Consuming one more unit today, $u'(c_1)$
- Saving one unit to get $1 + r$ units tomorrow; discounted utility per each such unit is $\beta u'(c_2)$

The Euler equation with log utility

Consider the case when $u(c_t) = \ln c_t$. This implies:

$$u'(c_t) = \frac{\mathrm{d}}{\mathrm{d}c_t}\left(\ln c_t\right) = \frac{1}{c_t} \tag{10}$$

The Euler equation, (9), under log utility:

$$\frac{1}{c_1} = \beta(1+r)\frac{1}{c_2}$$

Re-arranging:

$$\frac{c_2}{c_1} = \beta(1+r) \tag{11}$$

Solving the model

The IBC (3) and the Euler equation (11) determine the endogenous variables c_1 and c_2.

To solve for c_1 and c_2, substitute $c_2 = \beta(1+r)c_1$ from the Euler equation into the IBC:

$$c_1 + \frac{\beta(1+r)c_1}{1+r} = y_1 + \frac{y_2}{1+r}$$

Collecting terms:

$$(1+\beta)c_1 = y_1 + \frac{y_2}{1+r}$$

Solving for optimal consumption in period 1:

$$c_1 = \frac{1}{1+\beta}\left(y_1 + \frac{y_2}{1+r}\right) \tag{12}$$

Let $Y = y_1 + y_2/(1+r)$ denote the present value of life-time income. Then:

$$c_1 = \frac{1}{1+\beta}Y \tag{13}$$

and, since $c_2 = \beta(1+r)c_1$ in optimum:

$$c_2 = \frac{\beta(1+r)}{1+\beta}Y \tag{14}$$

Interpretation and implications

According to the neoclassical model:

- **Consumption smoothing:** Households want to smooth consumption over time
- **Permanent income hypothesis:** Consumption in any period is affected by (expected) *life-time* income
- Less patient households (smaller β) consume more today and less tomorrow
- Both income and substitution effect if interest rate changes (exercise?)

The desire to smooth consumption

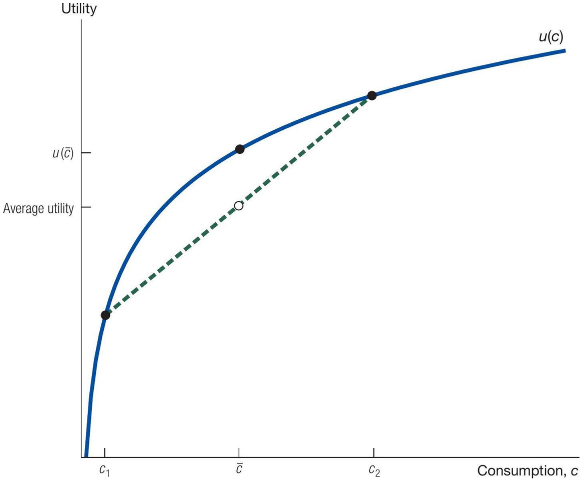


Figure 16.2 in Jones (2024)

Perfect consumption smoothing when $\beta(1+r)=1$

If the time-discount rate equals the interest rate, we have $\beta(1+r)=1$

Then:

$$c_1 = c_2 = \frac{1}{1+\beta}Y$$

Income will then be split equally between periods

Empirical evidence

Note the implications of the permanent income hypothesis:

- Suppose a *temporary* increase in income y_1 and many future time periods
- According to the PIH, most of the income increase should be saved. The **marginal propensity to consume, MPC** should be small

This neoclassical model is a good starting point for economic analysis

There are elements of consumption smoothing in the data

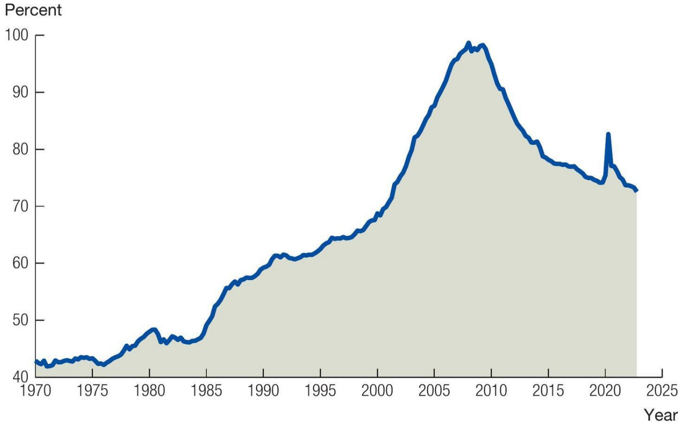
But some households are borrowing constained or less forward-looking than this model predicts

In Keynesian models (we will get there...), the MPC is assumed to be relatively high (but smaller than unity)

Consumption and asset prices

- Changes in asset prices (stock market, housing) important for the development of private consumption
- Since consumption constitutes such a large share of GDP, the economy becomes vulnerable to asset price fluctuations
- Risk of boom-bust cycles: sudden asset-price reversals tend to reinforce cyclical variations
- Should central banks try to counteract large swings in asset prices?

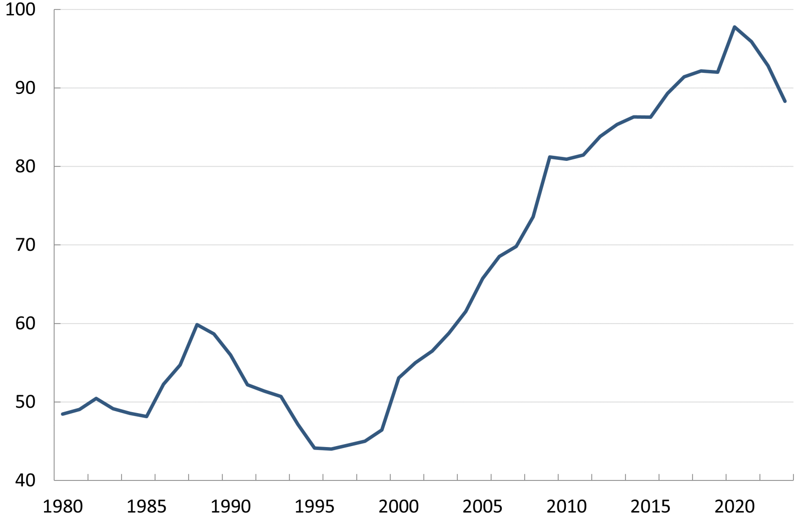
Household debt as a percent of GDP – U.S.



Source: The FRED database, *Flow of Funds of the United States*.

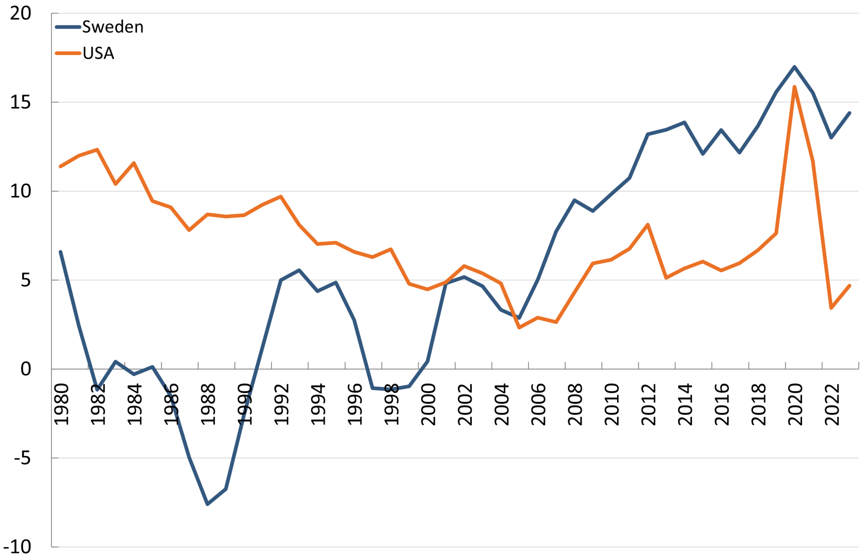
Figure 16.3 in Jones (2024)

Household debt as a percent of GDP – Sweden



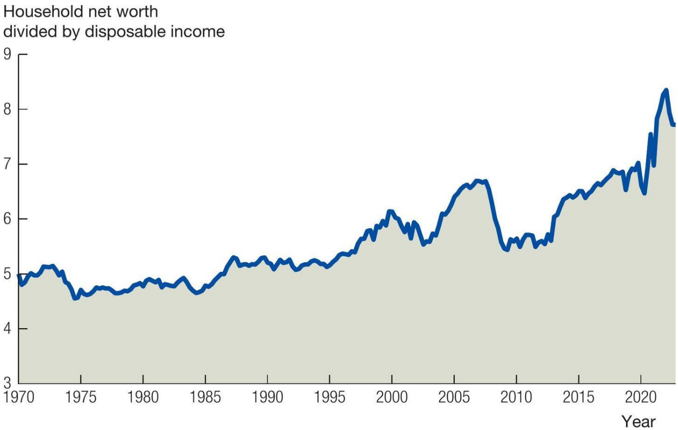
Source: Statistics Sweden

The household saving rate



Household savings, percent of disposable income. Source: OECD EO.

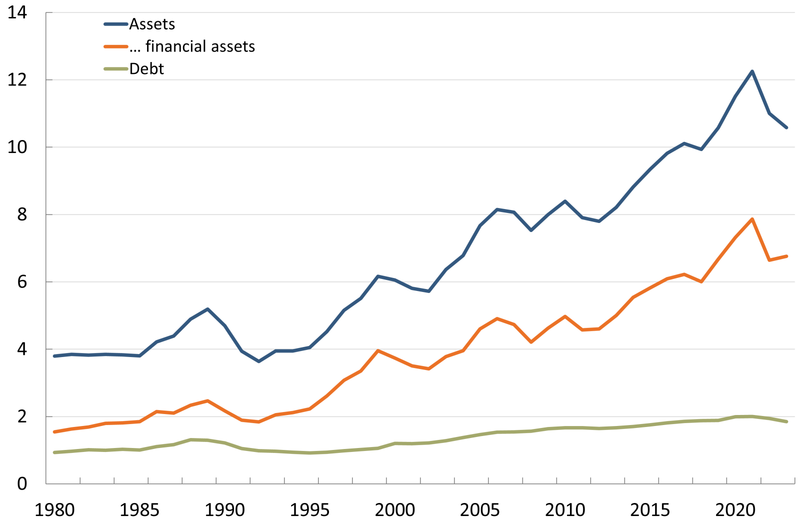
Household wealth (as a ratio to income) – U.S.



Source: Board of Governors of the Federal Reserve System (US), Households and nonprofit organizations; net worth as a percentage of disposable personal income, Level [HNONWPDPI], retrieved from FRED, Federal Reserve Bank of St. Louis; fred.stlouisfed.org/series/HNONWPDPI.

Figure 16.5 in Jones (2024)

Household assets and liabilities (as a ratio to income) – Sweden



Household assets and liabilities as a ratio to disposable income. Source: Statistics Sweden.

Investment

How do firms make investment decisions?

Suppose you (the firm) has one dollar. Invest in bank account or physical capital?

Bank account: 1 dollar today $\rightarrow 1 + r$ dollars tomorrow[1]

$$\text{Physical capital: 1 dollar today} \rightarrow \begin{cases} MPK \text{ production} \\ 1-\delta \text{ remaining capital} \\ \Delta p_k / p_k \text{ change in valuation of capital} \end{cases}$$

Equilibrium:

$$1 + r = MPK + (1-\delta) + \frac{\Delta p_k}{p_k} \tag{15}$$

[1] Jones uses R to denote the (real) interest rate whereas I use the more common r

User cost of capital

Reformulate (15) as:

$$MPK = \underbrace{r + \delta - \frac{\Delta p_k}{p_k}}_{\textbf{user cost of capital}}$$

If MPK is larger than the user cost of capital, the firm should invest more, etc

How much should a firm invest?

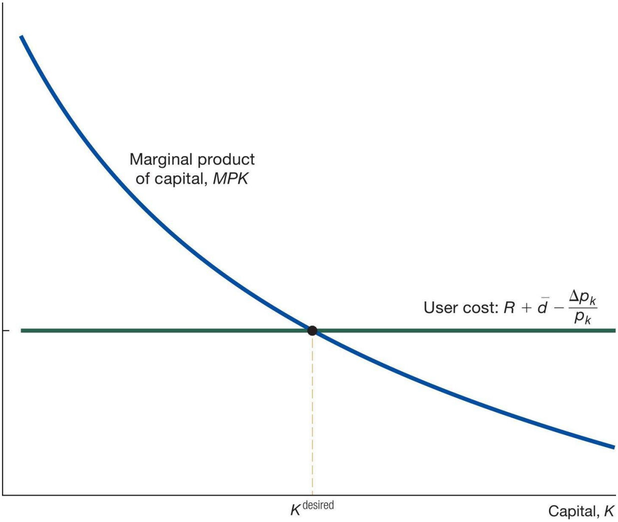


Figure 17.1 in Jones (2024)

Investment

A relationship like this will be important in the the next part of the course

We will assume that investment is:

... higher than normal when the interest rate is low relative to the MPK

... lower than normal when the interest rate is high relative to the MPK